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CSC580: Group 3: Final Exam

Final Exam Case – Electronic Document Management System Case Study

You have been entrusted with the expansion of the previous exam project creation of an Electronic Document Management System. This EDMS now aims to revolutionize document handling processes, enhance collaboration, and ensure information accessibility across the organization. The project presents unique challenges and requires innovative approaches to integrate cloud support and intelligent Optical Character Recognition (OCR) capabilities seamlessly.

The EDM system has several new (expanded) unique features that just be developed for the first time in such an application. These new features should include all list from the Midterm Exam and should ALSO include the following features:

1. The EDMS must integrate with cloud storage services such as Amazon S3, Google Drive, or Microsoft OneDrive to provide scalable and secure document storage.

2. It should support seamless uploading, downloading, and synchronization of documents between the local repository and cloud storage.

3. It should allow users to create custom metadata fields and define taxonomy structures for organizing documents effectively.

4. The system must enforce granular access control policies to regulate document access based on user roles, permissions, and security clearance levels.

5. The EDMS should integrate with intelligent OCR technologies to automatically extract text and data from scanned documents, images, and PDF files.

6. It should employ machine learning algorithms to enhance OCR accuracy, support multilingual recognition, and handle complex document layouts effectively.

7. It should provide collaboration tools such as document sharing, commenting, and real-time co-authoring to facilitate seamless collaboration among users.

8. It should support notifications, alerts, and reminders for document-related activities and deadlines.

CSC-580 Fall 2025

Final Exam

The Final Exam is to be done in groups just like the breakout sessions (Distance Learners may complete individually). All members' names must be put on each submission; however, each person is to submit a copy of the final exam on Canvas. The final page of the submission must have a team evaluation (1 or 2 paragraphs on the team's participation). All work must be submitted electronically, to Canvas, and everything must be in one file (word or pdf).

Every group must work alone. IF I FIND ANY ARTIFACTS (i.e., drawings, etc.) COMMON BETWEEN TWO GROUPS THEN BOTH GROUPS GET A ZERO (0) FOR THE FINAL EXAM GRADE.

***** NO LATE SUBMISSION WILL BE ACCEPTED FOR ANY REASON! *****

Every group must work alone. IF I FIND ANY ARTIFACTS (i.e. drawings, etc.) COMMON BETWEEN TWO GROUPS THEN BOTH GROUPS GET A ZERO (0) FOR THE MIDTERM GRADE.

1. Components: (15 pts)
    - **A. Document two candidate (2) components: The two components selected for detailed documentation are:**
        1. Cloud Storage Integration Component - Manages seamless synchronization with cloud services (Amazon S3, Google Drive, Microsoft OneDrive)
        2. OCR & Document Processing Component - Handles intelligent optical character recognition, text extraction, and multilingual support

    - **B. Define the component's purpose is and how it accomplishes its goals:**

1b) The Cloud Storage Integration Component serves as the bridge between the EDMS local repository and external cloud storage providers. It abstracts the complexity of managing multiple cloud services and provides a unified interface for document synchronization, upload, download, and backup operations.

Primary Responsibilities:

1. Cloud Service Abstraction — Hide the specific implementation details of different cloud providers (S3, Google Drive, OneDrive) behind a standardized interface
2. Bidirectional Synchronization — Maintain consistency between local and cloud repositories through smart sync mechanisms
3. Scalable Storage Management — Leverage cloud infrastructure to support unlimited document storage without local hardware constraints
4. Security & Encryption — Ensure documents are encrypted

during transmission and at rest in cloud storage

5. Error Handling & Retry Logic — Implement robust mechanisms to handle network failures, timeouts, and partial uploads
6. Metadata Management — Preserve document metadata during cloud transfers and maintain indexing for rapid retrieval

How It Accomplishes Goals: The component uses a Strategy Pattern to support multiple cloud providers through pluggable implementations. It implements:

1) CloudServiceAdapter interface — Standardizes operations across AWS S3, Google Drive, and Microsoft OneDrive
2) Synchronization Engine — Uses change detection (hash-based) to identify modified documents and sync only deltas
3) Connection Pooling — Maintains efficient connections to cloud services
4) Queue-Based Processing — Implements a task queue for asynchronous uploads/downloads to prevent blocking operations
5) Encryption/Decryption Pipeline — Applies AES-256 encryption before cloud transmission
6) Comprehensive Logging & Audit Trail — Tracks all cloud operations for compliance and debugging

2b) The OCR & Document Processing Component transforms scanned documents, images, and PDF files into machine-readable, searchable digital formats. It integrates intelligent optical character recognition technology with machine learning to enhance accuracy, support multilingual documents, and handle complex layouts effectively.

Primary Responsibilities:

1. Document Type Detection — Automatically identify document type, format, and content category
2. Intelligent OCR Processing — Extract text from images and scanned documents with high accuracy
3. Multilingual Support — Recognize and process documents in multiple languages (English, Spanish, Chinese, etc.)
4. Layout Preservation — Maintain document structure and formatting during text extraction
5. Machine Learning Enhancement — Use ML models to improve OCR accuracy over time
6. Data Validation & Correction — Detect and correct common OCR errors using dictionary and context

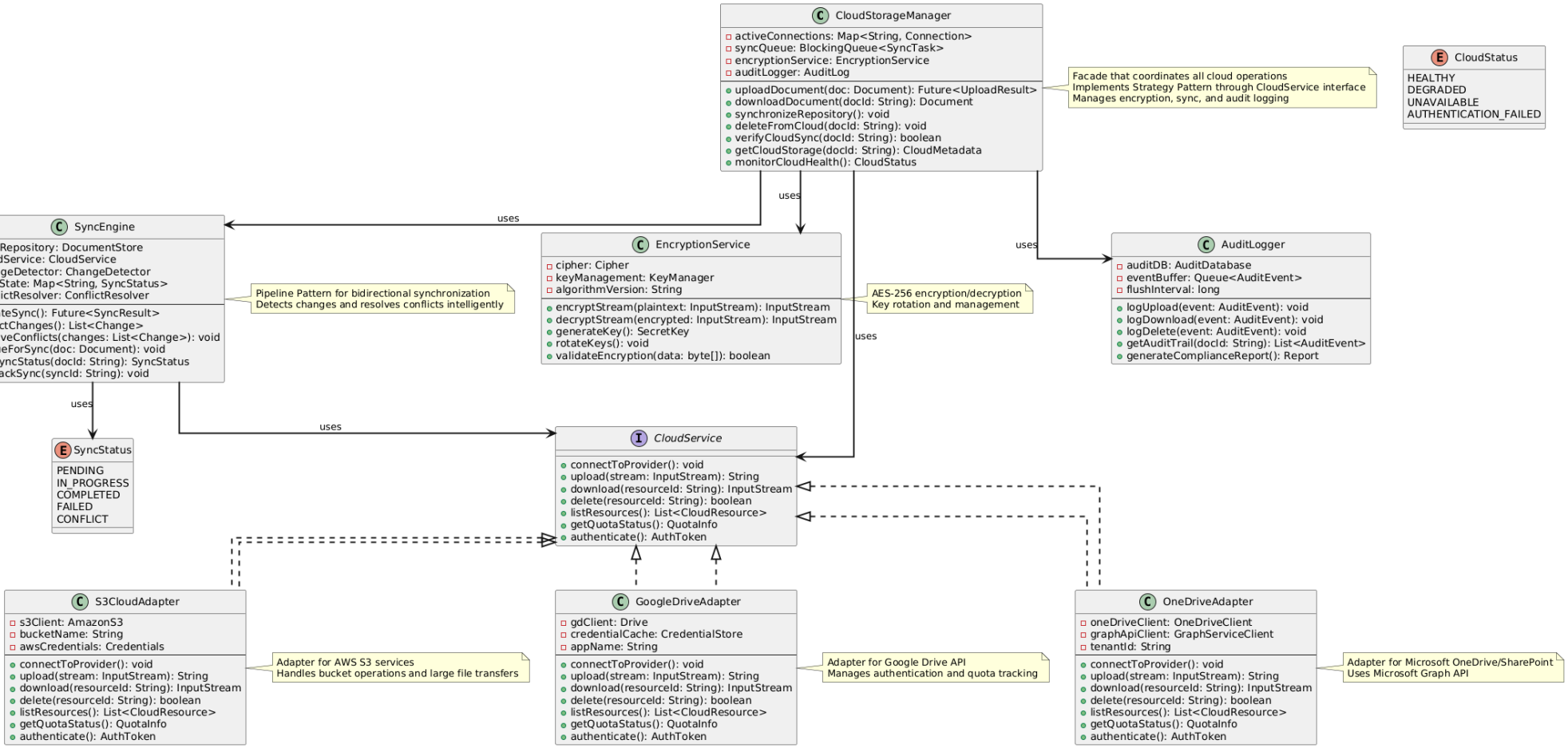
7. Metadata Generation — Automatically extract and generate metadata (date, author, keywords)
8. Quality Scoring — Assign confidence scores to extracted text and flag uncertain sections

How It Accomplishes Goals: The component implements a Pipeline Pattern with multiple processing stages:

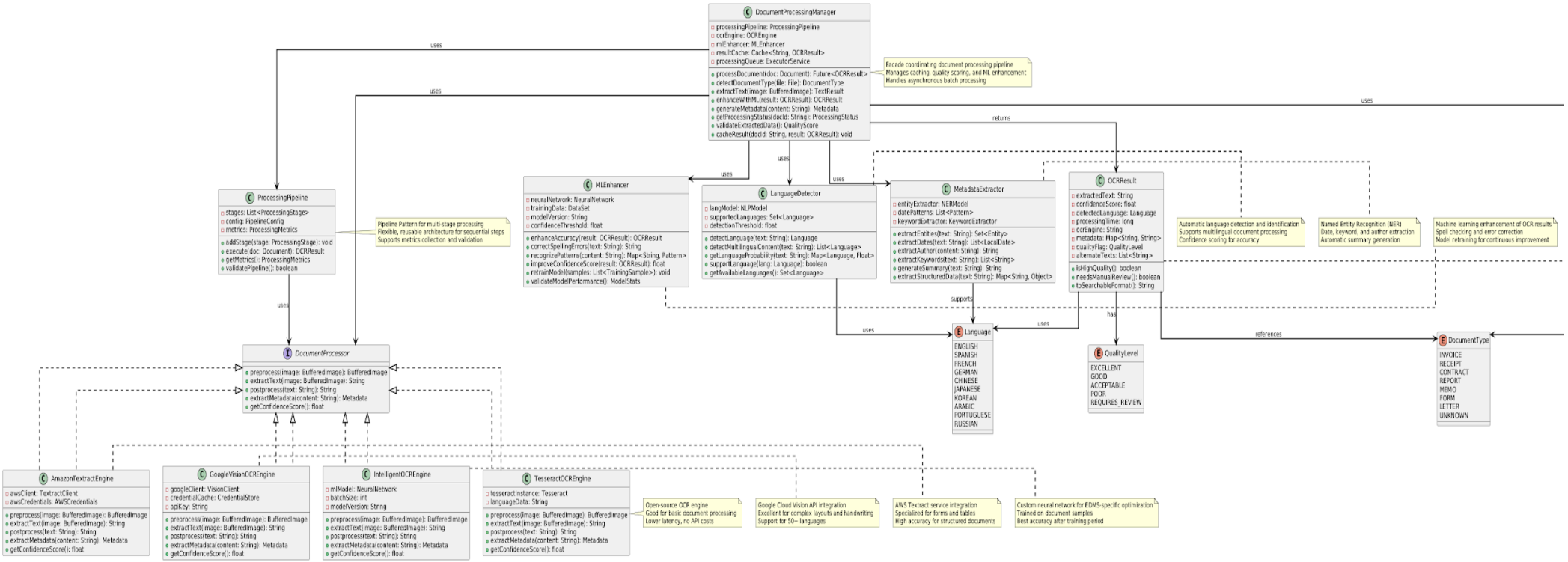
1) DocumentProcessor interface — Standardizes processing across different document types
2) Multi-Stage Pipeline — Preprocessing, OCR, post-processing, and metadata extraction
3) ML Model Integration — Leverages trained neural networks for improved accuracy
4) Confidence Scoring — Maintains quality metrics for extracted data
5) Caching Layer — Stores processed results to avoid reprocessing identical documents
6) Error Recovery — Implements fallback strategies for difficult sections
7) Asynchronous Processing — Handles large documents without blocking the system

C. Draw the component, including the classes that would be part of this component (be sure to identify the methods that would serve as the interfaces to this component also). You may use Internet resources as a starting point for the diagrams.

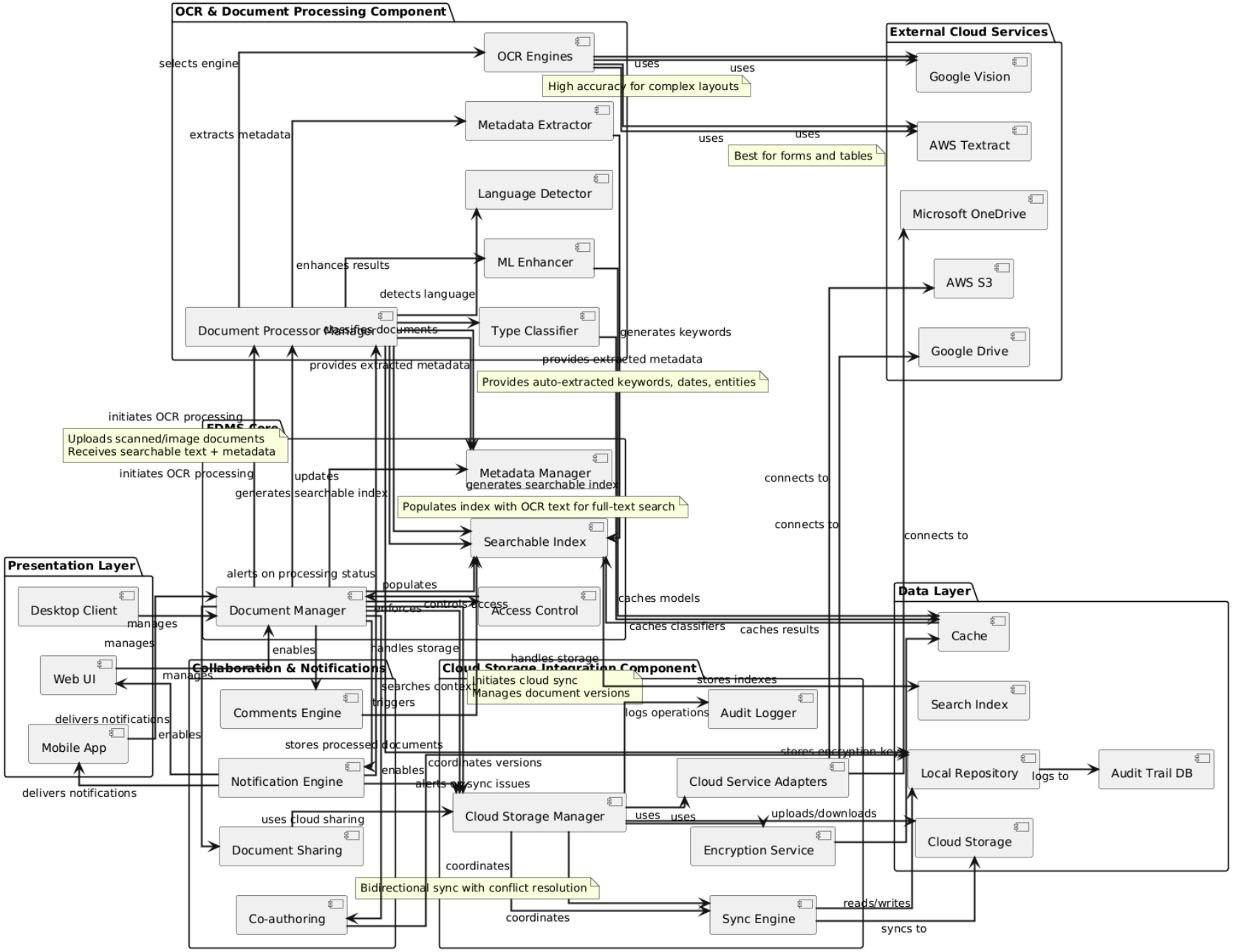
Component 1 Class Diagram



Component 2 Class Diagram



Integration Class Diagram:



2. Architecture: (25 pts.)
   a. Choose two (2) architectural styles that the team believes will be suitable for this application and draw them both. You may use Internet resources as a starting point for the diagram

Layered + Hexagonal (Port &Adapters)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│            Presentation Layer (Web/Mobile UIs, API Controllers, BFF)          │
└──────────────────────────────────────────────────────────────────────────────┘
                  ↑
                  | application calls / DTOs
┌──────────────────────────────────────────────────────────────────────────────┐
│   Domain Core (Hexagon): Use Cases · Entities · Policies                     │
│   Ports (interfaces):                                                        │
│   - Identity & ConsentPort       - Orders/FHIRPort      - NotificationPort   │
└──────────────────────────────────────────────────────────────────────────────┘
         ↑                  ↑                  ↑
         | adapters         | adapters         | adapters
┌──────────────────────────────────────────────────────────────────────────────┐
│   Infrastructure Adapters:                                                   │
│     - OIDC/SSO IdP Client    - EHR/FHIR Adapter     - Email/SMS Provider     │
│     - Database (Consent, Orders, Audit)  - Object Storage - Message Bus      │
└──────────────────────────────────────────────────────────────────────────────┘
```

Layered +Hexagonal (Ports & Adaptors)

- **Why suitable:** separates domain (consent, orders) from delivery (HTTP/UI) and infrastructure (EHR, DB). Ports define what the domain needs, adapters implement it (e.g., FHIR adapter, IdP adapter).
- **Strength to Enterprise:** Easier swaps (change EHR vendor, switch IdP) without touching core logic; high testability with port mocks, clear audit and policy seams.

Microservices + Event-Driven Backbone

```
┌──────────────────────────────── API Gateway / BFF ────────────────────────────────┐
│ (Routes Web/Mobile/Partner traffic to services: auth, rate limit)                 │
└───────────────────────────────────────────────────────────────────────────────────┘
          │                                  │                                  │
          ▼                                  ▼                                  ▼
┌──────────────────────────┐   ┌──────────────────────────┐   ┌──────────────────────────┐
│ Identity & Consent Service │   │ Orders / FHIR Service    │   │ Notifications Svc        │
│ (REST/gRPC + own DB)       │   │ (REST/gRPC + own DB)     │   │ (REST + own DB)          │
└──────────────────────────┘   └──────────────────────────┘   └──────────────────────────┘
          │ publish/subscribe          │ publish/subscribe          │ subscribe
          ▼                            ▼                            ▼
┌──────────────────────────────────────────── Event Backbone ────────────────────────────────────────────┐
│  topics: OrderCreated · ConsentRevoked · NotificationRequested                                         │
└────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

b. Fill out the following table for your candidate architectures to determine if one, or both, styles you have selected could be a good choice.

Attribute	**Importance Factor (1-5, 1 = most importance**	**Why is this attribute important?**	**Met by this Architecture ? (Y/N)**	**How does the architecture meet it?**
Usability	4	clinicians/patients need a clear, consistent experience to reduce errors and support adoption	Y	Presentation layer is isolated from the domain; BFF/API controllers provide consistent contracts; enables UX iteration, A/B tests and accessibility checks without breaking core

				logic.
Reliability	2	Health data workflows must work first time, every time to avoid delayed care.	Y	Uses cases in the domain layer are determined and idempotent; adapters add retries/timeouts; error handling and validation centralized; automated tests around ports.
Performance	3	Snappy responses improve clinician throughout and patient satisfaction.	Y	Controllers map to slim use cases; caching and batching live in adapters; asynchronous messaging available via the message-bus adapter for non-critical paths.
Supportability	3	Ops and help-desk need clear diagnostics to resolve issues quickly.	Y	Clean seams (ports) allow structured logging, tracing and health checks per adapter; failures are localized (e.g., EHR

				adapter) with targeted dashboards and runbooks.
Security	1	PHI must be protected (HIPPA/HITECH); breaches are unacceptable.	Y	Central policy enforcement in domain; OIDC/SSO adapter, least-privilege scopes; encryption and audit logging handled in Infra adapters; secrets isolated from app code.
Safety	1	Clinical safety requires correct rules and guardrails to prevent harm.	Clinical rules/validators live in the domain (not UI); strong input validation on ports; feature flags and canary release reduce risk of unsafe changes.	
Availability	2	Service must be reachable during clinic hours and for patients after	Y	Stateless presentation tier can scale horizontally; message-bus

		hours.		adapter supports graceful degradation for async tasks; fallbacks/circuit breakers in adapters.
Maintainability	2-3	Requirements change often; code must be easy to modify safely.	Y	Hexagonal design isolates business logic from vendors; swap adapters (EHR, Idp, SMS) without touching core; high unit testability of use cases; CI/CD enforces quality gates.

c. Create a narrative response to answer in depth: 1) how the architecture styles were selected, 2) why they were selected (what purpose they should fulfill), 3) how they strengthen the enterprise environment (functionality it adds), and 4) which alternatives were evaluated.

How the Architecture Styles were Selected

We filtered candidate styles using the course's emphasis on clear interfaces, modularity and traceability (ports, contracts and clean seams). That les us first to Layered+Hexagonal for its crisp boundaries between UI, domain and infrastructure. Because the problem also implies integration with external systems (e.g., EHR/FHIR, identity providers) and possible

growth, we added a second style Microservices with an event backbone that cleanly separates concerns into deployable services and uses event to decouple workflows, consistent with the lecture guidance on maintainable interactions and sequence-of-operations clarity.

Why they were selected (purpose they fulfill)

- **Layered + Hexagonal** is ideal to deliver a fast, testable MVP where the domain model (consent, orders) is protected behind ports. Any vendor/EHR/IdP dependence is pushed into adapters, so we can change providers without rewriting core logic.
- Microservices + Event-Driven serves the purpose of scale and autonomy: identity/consent, orders/FHIR and notifications evolve independently with topics (e.g., *OrderCreated,ConsentRevoked*) enabling loose coupling and asynchronous processing.

How they strengthen the enterprise environment

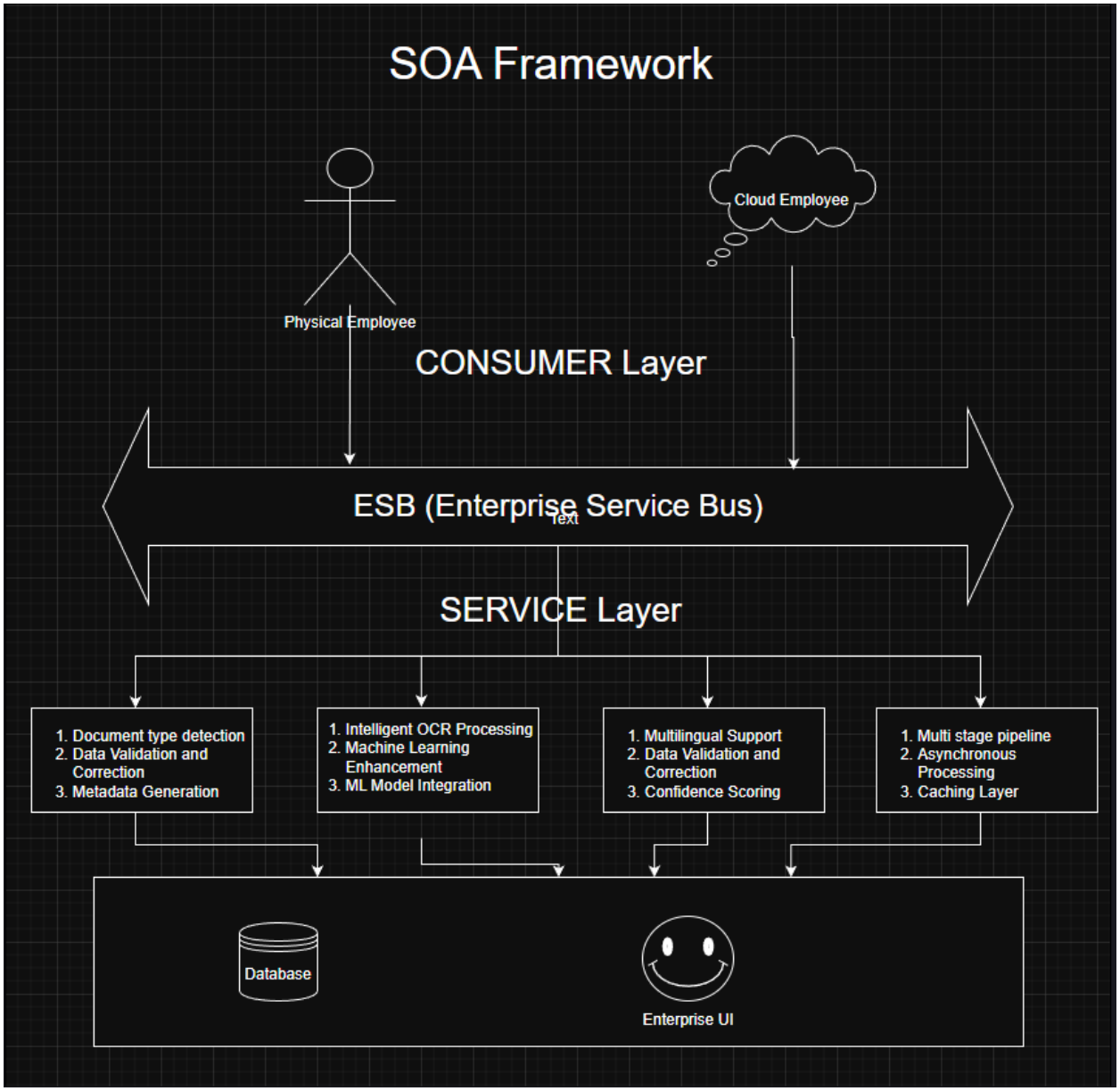
- Layered + Hexagonal: strengthens compliance and auditability: one CI/CD pipeline, a unified audit trail, and single-place policy enforcement. It's also cost-efficient operationally: fewer moving parts, simpler on-call.
- Microservices + Event-Driven: strengthens resilience and scalability: per-service horizontal scaling, circuit breakers at service edges, and observability (per-service dashboards). Events naturally feed analytics and reporting without coupling runtime paths.

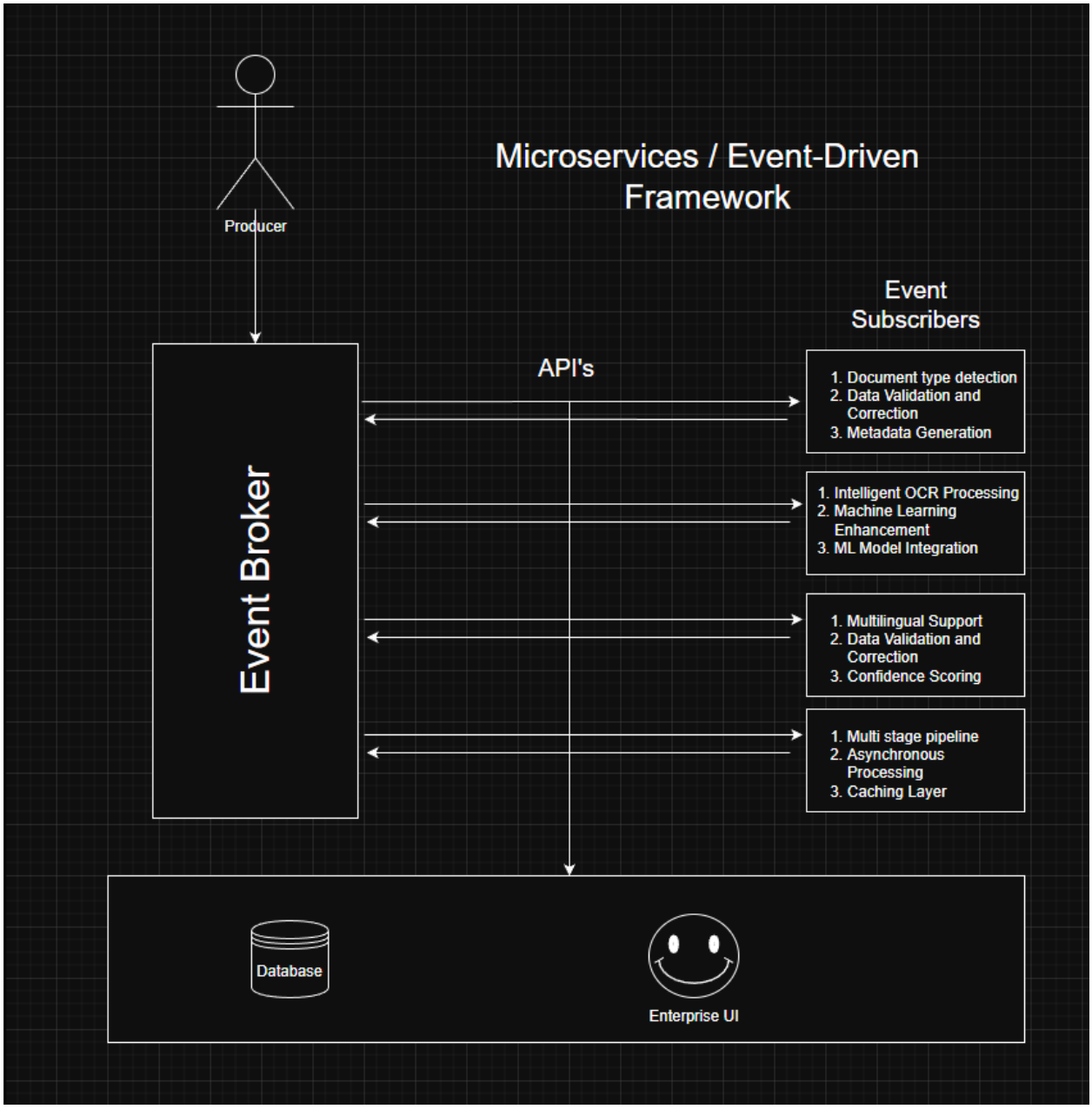
Alternatives evaluated

- Classic 3-Tier (UI_Logic-DB). Simple but couples domain to a specific DB and blurs external boundaries; weaker for swapping vendors and for unit testing core logic.
- SOA with a heavy ESB. Central breaker simplifies writing but becomes a bottleneck and creates organizational coupling.
- Serverless-first. Attractive for bursty traffic, but cold starts and distributed debugging add friction; difficult multi-function tracing for regulated audits.
- Event-only monolith. Gains decoupling, but without true service boundaries you still hit monolith scaling limits.

3. Framework: (25 pts.)
    a. Choose two (2) Enterprise Framework styles that you feel would be suitable for this application and draw them both. You may use Internet

resources as a starting point for the diagrams.




Microservices / Event-Driven Framework
Producer
Event Broker
API's
Event Subscribers
1. Document type detection
2. Data Validation and Correction
3. Metadata Generation
1. Intelligent OCR Processing
2. Machine Learning Enhancement
3. ML Model Integration
1. Multilingual Support
2. Data Validation and Correction
3. Confidence Scoring
1. Multi stage pipeline
2. Asynchronous Processing
3. Caching Layer
Database
Enterprise UI

b.

Fill out the following table for your candidate frameworks to determine if one, or both, frameworks you have selected could be a good choice.

Service Oriented Architecture Framework				
Attribute	**Importance Factor 1-8 1 being *MOST* important**	**Why is this attribute important?**	**Met by this Framework? (Y/N)**	**How does the Framework meet it?**
Usability	1	Multiple users with varying skill levels throughout entire companies will be using this product. If the product is not highly usable then critical business information may be lossed or not captured due to technological gaps in different areas of the company.	y	since SOA allows multiple different areas of the software to be built independently and be loosely coupled with other areas of the software it is feasible to develop or purchase a standalone UI that is particulary suited to be very user friendly
Reliability	7	Reliability is not top priority. Similar to performance, it is not going to be a show stopper if this system doesn't manage all tasks as precisely as we would like in all situations. Either way, the document is generally getting stored somewhere with some level of taxonomy and can be accessed eventually regardless of how reliable the extra features were.	n	SOA is more modern and relies a lot on cloud based services. Sometimes things may be dropped in the process of transferring information. Loss of connection may come into play as well. Many SOA frameworks use an Enterprise Service Bus as well to route the consumer to the appropriate service. If this is down the whole system is down
Performance	6	High perfomance is not critical for this service. While it is important to be able to collect and store documents effectively, it is almost never important to be able to access them at light speed. With that being said, poor performance can lead to frustration in users causing unnecessary friction in organizations.	n	Although more agile than traditional frameworks in a sense that each service does not need to wait for input from other functions in the system it is still at the mercy of connection speed and queues when using cloud based solutions.
Supportability	4	Since all users in the company, with varying level of skill, will be using this documentation system, it is highly important that ample support is available. While on demand face to face interaction is preferred, products of this caliber tend to do well with FAQ's and online help menu's as well.	y	Services can be purchased directly from suppliers off the shelf who specialize in that particular area. Contracts can be developed independently of other areas of the software which can allow for any level of support needed across multiple areas of the software.
Security	2	This product will hold highly sensitive business information and quite possibly all documented information for the business itself. If the information isn't secure, competitors may be able to extract trade secret information or company information may go public prematurely leading to loss of competitive edge.	y	Generally in SOA single instances of data transactions are not kept around for people to discover. Typically the service just performs the transaction and returns the result to the user. For expample, a user creates a document and commits it to the database tagged with metadata. A commit Service may process the service of processing the document to the database, but ultimately the only record of the transactions would be housed locally in the database.
Safety	8	This product will have no direct interaction physically with employees therefore there should be many if at all safety concerns with using this product.	n	SOA does not look like it lends itself to anything that would involve or prevent bodily harm to anyone. Nor is it responsible for operating or monitoring equipment that may do harm as well.
Availability	5	Odds are most of the document transfers and production will be during business hours, this will allow for ample time on off hours to complete maintenance activities and take the system down for fixes. Standard maintenance activities could be outsourced to suppliers in different countries to offset cost and take advantage of time difference.	y	SOA has full capability to be cloud based. These services could have activities outsourced at any given time. Also cloud applications tend to be highly available given the large corporations that run the Services.
Maintainability	3	This product will almost certainly become a legacy software in the future. If a company dedicates document storage solely to one system with standardized filing and taxonomy it will be very challenging for them to be able to switch systems in the future. IT Personnell will have to maintain the software for many years after implementation.	y	SOA has the ability to advance in all areas independently. Growth in one area will not be stunted because it is highly dependent on another. With that being said, as the system grows larger and some areas are still loosely coupled with others, these couplings will become more and more complex and intricate.

Microservices / Event-Driven Framework				
Attribute	**Importance Factor 1-8 1 being *MOST* important**	**Why is this attribute important?**	**Met by this Framework? (Y/N)**	**How does the Framework meet it?**
Usability	1	Multiple users with varying skill levels throughout entire companies will be using this product. If the product is not highly usable then critical business information may be lossed or not captured due to technological gaps in different areas of the company.	y	since MSE allows multiple different areas of the software to be built independently and be loosely coupled with other areas of the software it is feasible to develop or purchase a standalone UI that is particulary suited to be very user friendly
Reliability	7	Reliability is not top priority. Similar to performance, it is not going to be a show stopper if this system doesn't manage all tasks as precisely as we would like in all situations. Either way, the document is generally getting stored somewhere with some level of taxonomy and can be accessed eventually regardless of how reliable the extra features were.	y	MSE is more modern and relies a lot on cloud based services. Sometimes things may be dropped in the process of transferring information. Loss of connection may come into play as well. It does not however lend itself to a ESB like SOA.
Performance	6	High perfomance is not critical for this service. While it is important to be able to collect and store documents effectively, it is almost never important to be able to access them at light speed. With that being said, poor performance can lead to frustration in users causing unnecessary friction in organizations.	n	Although more agile than traditional frameworks in a sense that each service does not need to wait for input from other functions in the system it is still at the mercy of connection speed and queues when using cloud based solutions. Also, each service can be kicked off simultaneously.
Supportability	4	Since all users in the company, with varying level of skill, will be using this documentation system, it is highly important that ample support is available. While on demand face to face interaction is preferred, products of this caliber tend to do well with FAQ's and online help menu's as well.	y	Services can be purchased directly from suppliers off the shelf who specialize in that particular area. Contracts can be developed independently of other areas of the software which can allow for any level of support needed across multiple areas of the software.
Security	2	This product will hold highly sensitive business information and quite possibly all documented information for the business itself. If the information isn't secure, competitors may be able to extract trade secret information or company information may go public prematurely leading to loss of competitive edge.	y	Generally in MSE single instances of data transactions are not kept around for people to discover. Typically the service just performs the transaction and returns the result to the user. For expample, a user creates a document and commits it to the database tagged with metadata. A commit Service may process the service of processing the document to the database, but ultimately the only record of the transactions would be housed locally in the database.
Safety	8	This product will have no direct interaction physically with employees therefore there should be many if at all safety concerns with using this product.	n	MSE does not look like it lends itself to anything that would involve or prevent bodily harm to anyone. Nor is it responsible for operating or monitoring equipment that may do harm as well.
Availability	5	Odds are most of the document transfers and production will be during business hours, this will allow for ample time on off hours to complete maintenance activities and take the system down for fixes. Standard maintenance activities could be outsourced to suppliers in different countries to offset cost and take advantage of time difference.	y	MSE has full capability to be cloud based. These services could have activities outsourced at any given time. Also cloud applications tend to be highly available given the large corporations that run the Services.
Maintainability	3	This product will almost certainly become a legacy software in the future. If a company dedicates document storage solely to one system with standardized filing and taxonomy it will be very challenging for them to be able to switch systems in the future. IT Personnell will have to maintain the software for many years after implementation.	y	MSE has the ability to advance in all areas independently. Growth in one area will not be stunted because it is highly dependent on another. With that being said, as the system grows larger and some areas are still loosely coupled with others, these couplings will become more and more complex and intricate.

The Microservices / Event-Driven Framework was selected for our project. MSE has evolved from SOA and just provides a more fluid option that fits the dynamic solution we require. The biggest deciding factor is MSE's lack of an ESB. Ultimately MSE achieves higher reliability while meeting or exceeding SOA in every other category. Also MSE utilizes lightweight API's which are a more modern request system that more closely follows market trends and allows for the most diverse solutions.

c. Create a narrative response to answer in depth: 1) how the frameworks were selected, 2) why they were selected (what purpose they should fulfill), 3) how they strengthen the enterprise environment (functionality it adds), and 4) which alternatives were evaluated.

The SOA and MSE frameworks were selected because they are highly effective at leveraging solutions that have already been produced by companies that are highly skilled and proven in any one area. For example, Document Validation, Document Type Detection, and MetaData Generation already have significant suppliers in this space with cloud based solutions that are readily available and scalable for any size operation. Also, with these frameworks it is possible to search outside to multiple competitive suppliers in different markets rather than one size fits all solution.

Because we are able to develop or purchase proven solutions for each specific environment of the software individually we can also drive down the overall cost of the software enterprise by increased competition. Since different aspects are not very coupled, and requests are handled independently we can search for the best suppliers in each space and have them compete with each other to achieve the best prices on each individual component.

Finally, these frameworks allow for innovation and advancement in different areas without affecting the enterprise as a whole. For example, our enterprise will rely on multiple AI and ML initiatives such as Intelligent OCR processing and ML Model integration. These areas are constantly evolving and would require regular updates and constant revamping. In SOA and MSE it is possible to take advantage of all of these advancements independently as they are not directly coupled or influenced by other areas of the software. If another company significantly surpasses a current supplier in the market we could easily switch to them and have them handle that portion of the business while keeping other suppliers intact in other areas. Each area can be scaled and advanced as slow or as fast as necessary without having to rework the entire system due to taking advantage of one area's updates.

4. Design Patterns: (15 pts.)

a. Choose at least five (5) patterns that would be suitable to your system and describe:

i. How they were selected

ii. How they will be useful

iii. How they will be implemented

iv. How it will (or will not) be reused throughout the enterprise

Answer:

1) For the Factory Method pattern, the team selected it because the EDMS must support multiple cloud providers (Amazon S3, Google Drive, OneDrive) behind a unified “cloud storage” abstraction, and hard-coding `if/else` object creation would be brittle and hard to extend.
    a) It is useful because it encapsulates the creation of `CloudServiceAdapter` implementations, making it easy to add or swap providers without touching business logic, while also centralizing configuration, retries, and security concerns.
    b) Implementation-wise, the design introduces a `CloudServiceAdapter` interface plus concrete adapters (e.g., `S3CloudAdapter`, `GoogleDriveAdapter`), with a `CloudServiceFactory` (and possibly provider-specific factories) responsible for instantiating the right adapter based on configuration or user choice.
    c) This factory layer can be packaged as a shared enterprise library so any other application that needs uniform access to these same cloud services can reuse the same adapters and factories with minimal change.
2) The Builder pattern was chosen because EDMS documents are complex aggregates combining core metadata, taxonomy, OCR results, confidence scores, and access control, and telescoping constructors would be unreadable and error-prone.
    a) It is useful in this system for creating `Document` objects with different combinations of optional attributes (e.g., scanned vs. born-digital documents) while enforcing validation rules in one place before the object is constructed.
    b) Implementation will use a `DocumentBuilder` with fluent methods such as `withTitle`, `withSecurityLevel`, and `withOcrData`, plus a `build()` method that validates mandatory fields and then calls a private `Document` constructor; optional “director” classes can encapsulate standard recipes for common document types like “ScannedInvoice” or “TechnicalDrawing”.
    c) Across the enterprise, the same Builder pattern (and even shared base builder/validation utilities) can be reused for other complex aggregates such as user profiles, workflow definitions, or report specifications, even if the concrete `DocumentBuilder` itself is domain-specific to EDMS.
3) The Singleton pattern was selected after identifying components that must exist exactly once per application instance, such as the configuration manager, the global cloud sync queue coordinator, and centralized logging/audit services.
    a) It is useful here because it guarantees a single, globally accessible instance for expensive or consistency-critical resources, preventing configuration drift, duplicated queues, or fragmented audit logs, while enabling lazy initialization for heavy components such as the OCR

engine pool.

b) Implementation will use thread-safe Singleton idioms (e.g., double-checked locking for `ConfigurationManager`, eager initialization for logging) that expose `getInstance()` methods and encapsulate loading configuration, managing worker pools, or writing audit entries.

c) At the enterprise level, the pattern's techniques (thread-safe singletons for config and logging) are reusable, but individual Singletons tend to be application-specific, and in many newer services, a dependency-injection container may be preferred over Singletons for most shared services.

4) The Bridge pattern was chosen because the EDMS has two independent axes of variability: document abstractions (text, spreadsheets, images, drawings) and storage/processing implementations (local filesystem, S3, Google Drive, OCR-enhanced backends), and naïve inheritance would cause a class explosion.

   a) It is useful by decoupling the `Document` abstraction hierarchy from the `StorageImplementor` hierarchy so new document types or new storage providers can be added independently, and documents can switch storage strategies (e.g., migrate from local to S3) at runtime without changing their core logic.

   b) Implementation will define an abstract `Document` that composes a `StorageImplementor` interface; concrete documents override operations like `save` and `render`, while storage implementors (e.g., `LocalFileStorage`, `S3CloudStorage`, `GoogleDriveStorage`) handle the actual read/write semantics for each backend.

   c) This separation is highly reusable across the enterprise for other "two-dimension" problems (e.g., business entities vs. data sources, UI widgets vs. platforms), and the pattern can be standardized as a recommended approach wherever orthogonal dimensions would otherwise multiply classes.

5) The Decorator pattern was selected because EDMS documents may need optional, combinable behaviors: encryption, compression, watermarking, audit logging, digital signatures; Without bloating the base `Document` class or creating many specialized subclasses.

   a) It is useful because it lets the system wrap a core `DocumentComponent` with decorators such as `EncryptionDecorator`, `CompressionDecorator`, and `AuditLogDecorator` at runtime, composing exactly the feature set needed per document type or security policy while keeping each concern in its own class.

   b) Implementation will define a `DocumentComponent` interface, a `BasicDocument` concrete component, an abstract `DocumentDecorator` that delegates to a wrapped component, and

concrete decorators that override methods like `getContent`, `setContent`, or `save` to add behavior before or after delegating.

c) Since many other enterprise services also need pluggable behaviors (e.g., request logging, caching, security checks, compression), the same Decorator approach and even shared generic decorators (logging, metrics, caching) can be reused in other applications, making this pattern one of the most broadly reusable choices across the organization.

5. Testing: (20 pts.)
   a. Create a Test Plan - Describe in detail how the application will be tested – this will include:
      i. Which testing scenarios will be utilized (regression and/or standard scenarios)
      ii. Testing Roles
      iii. Quality Assurance Vs. User Acceptance Scope
      iv. Example Scenarios and Scripts
      v. What will constitute successful tests and overall testing success

******A complete testing plan should be at least 10 pages in length with no obvious gaps in the test plan and the included test scenarios. Please focus on including business and technical testing strategies (including specialized testing covered in lecture), regression testing, unit testing, TDD, etc.******

CSC 580
(Implementing cloud storage and OCR)
(Phase #1)
Detailed Test Plan

1.General Information

System or Project ID/Acronym:	CSOCR-1	**Creation Date:**	12/05/2028
Client Agency:		**Modification Date:**	N/A
Author(s):	Michael Kelbert, Joseph Mendoza, Bradley Moore, Rafique Agyare	**DTMB Authorized by:**	

2.Privacy Information

This document may contain information of a sensitive nature. This information should not be given to persons other than those who are involved with this system/project or who will become involved during its lifecycle.

3.Revision History

The Project Manager will maintain this information and provide updates as required. All updates to the Project Management Plan and component plans should be documented in this section.

Revision Date	Author	Section(s)	Summary

10/01/2025	PM	Item in scope	Decision was made to introduce Multilingual support within OCR machine learning model to help with non-english document categorization
11/01/2025	PM	Item in scope	Backend Service provide has been decided as Amazon S3

4.Overview

The Detailed Test Plan supports the test type details outlined in the project's Test Strategy document. It details information for specific test phases (waterfall), or for an occurrence of Deploy Product Increment (agile) testing. The project's Test Strategy document can be referenced to satisfy items in this plan.

Whenever the words "test phase" are used within this document this is in reference to either a specific test phase (waterfall) or for an occurrence of Deploy Product Increment (agile) testing.

Projects should complete the sections below or insert a link to other project documents containing the information.

5.Scope

Document the scope of what will be tested during this phase of testing. A list generated from the project's testing tool can be inserted or a link added in lieu of completing the table.

5.1Items in Scope

Inputs to Testing (i.e. Module, Interface or Process Name/No.)	Functionalities/Features	Test Type(s)
Cloud upload	Files move from local storage to cloud storage - Shows a catalog of all uploaded files	**White box**
Object Character recognition model	When documents are scanned they are processed through a OCR model to extract handwritten and typed text into a solely text document	**White Box**

5.2Items Out of Scope

Inputs to Testing (i.e. Module, Interface or Process Name/No.)	Functionalities/Features
Existing document environment	Acts as client for all active users
Profile Creation and permissions control	Permission controls will still be accessible to managers and permissions will apply to local and cloud files

5.3Test Scenarios

Test Scenario Number	Description of Scenario
1	Creation and editing uploads document to cloud
2	Scanning document triggers automatic OCR process

5.4Test Cases

Test Case Number	Test Case Description
1	The user will create a document. They will be able to see the document in the local environment. Working with the Dev they should be able to confirm that the document enters the queue for cloud upload. After cloud upload the user is able to re-enter the document save changes and document will load into cloud upload queue
2	The user will scan in a document with both english and non-english text. OCR will run once the document is scanned and a transcript will be

	created of the document. The user will check the accuracy of the scan. The user will then edit the document save and the document will be saved to the cloud.

6.Test Phase Management

Please provide answers to the requested information below (or attach a link to a document or tool that contains the information).

6.1Objectives

Describe the business and technical goals of this test phase.

The business goal of this project is to have secure backups of all documents created throughout the company, while still maintaining user security roles. The implementation of OCR will allow for ease of search for scanned documents and ease of workflow. The Technical goal for this project is to create a cloud storage for all documents created by employees. A OCR machine learning model will also be created to extract information off of scanned in documents

6.2Requirements Traceability

Provide links to tools that will contain the Requirements, Test Cases and Traceability, or describe how traceability will be established between Requirements and Test Cases. Insert links to the locations (SharePoint or tool) where these are stored.

Provided above

6.3Data Management

Provide information on how often (and the process involved) the test data will be refreshed, backed up and restored etc., so that effective test planning and execution can take place.

Cloud backup and OCR will happen immediately either after document is created/edited or when a document is scanned into system

6.4Pass Strategy

Document how tests will be executed in order to obtain a passed status (i.e. iteratively through multiple repetitions).

Test will be executed in an iterative process. Each step will grow on top of each other starting with simple document creation all the way until the user is able to scan in a document, receive OCR transcript then edit the transcript and upload is successful.

6.5Cycle/Iterations/Product Increment

Enter specific test types and tasks (i.e. Integration Testing, Regression Testing, batch processing, month-end reconciliation etc.) that are to be completed during each test cycle/iteration or product increment.

Testing Types/Tasks	Start Date for each Test Cycle/Increment/Product Increment	End Date for each Test Cycle/Increment/Product Increment
Integration testing (Connecting local environment to cloud environment)	01/01/2026	01/15/2026
Regression testing (Local environment still functions before changes were implemented Users can Create, Edit, Share, & Collaborate)	01/15/2026	01/29/2026

Batch Processing (Multiple files are created for stress testing upload queue)	01/29/2026	02/12/2026

6.6 <Test Phase> Entry Criteria

Provide a list of factors/conditions that must be met before beginning the test execution in this phase.

Backend process must be set up before testing is able to begin. As local storage should not change from existing functionality. Once background services like OCR and cloud storage connection is established the user will be able to start testing

6.7<Test Phase> Exit Criteria

Provide a list of factors/conditions that must be met before closing out this test execution in this phase.

User must be able to Create a document, have it saved to the cloud then pull the document for editing and have new changes saved to the cloud. The user must also be able to scan in a document with English and Non-English text and OCR must be able to make a transcript of that text

7.Test Controls

Test Controls are used to monitor the test phase and provide status to the project.

7.1Key Dates and Deliverables

List all project deliverables and associated tasks including the information described in the tables below (or attach a list of tasks from the Enterprise Project Tracking Tool or any other tracking tool that contains these details).

Deliverable	Due Date	Responsible	Reviewer	Review Type
Cloud provider blackbox testing	12/25/25	Cloud service provider	IT Developer/ IT management	
Test plan	10/31/2025`	Quality Assurance tester	All Stakeholders	
Internal black box testing results	12/25/25	IT Developers	IT Management	

Refer to the Organizational Project Management Operating Procedures and the project's Project Management Plan for the project for information on testing deliverables, review type and appropriate reviewers. Note any changes here.

7.2Test Milestones

Include milestones related to test planning, data preparation, automated testing, environment readiness, etc. for this test phase.

Test Phase Milestones	Start Date	End Date
	09/01/2025	09/15/2025

Backend integration setup		
OCR machine learning model testing	09/16/2025	09/24/2025
OCR Machine learning model deployment	09/25/2025	10/10/2025
User environment readiness	10/10/2025	10/31/2025

7.3Change Request Management

If the project will be using procedures different from the procedures listed in the project's Project Management Plan (PMP) then list the differences here.

N/A - will be using PMP

7.4Defect Tracking and Defect Management

Describe how defects will be tracked and managed during this testing phase. An example is provided as a starting point in Appendix A – Example Defect Management Process for Waterfall Projects.

If an issue is discovered a Bug Ticket must be opened and assigned to the developer. If an issue is Critical it will be discussed in meeting and possible solutions will be brainstormed with all stakeholder

7.5Defect SLA for Resolution

Include a link to the project's Test Strategy document or list each category of defect and the timeframes for identifying cause/fix and when retesting should occur.

Defect Severity	Timeframe for Analysis and Correction
Blocker	Immediate acknowledgement and analysis must be started ASAP. Correction is expected within <1 Day
Critical	Immediate Acknowledgement and analysis must be started the same day. Correction is expected within 1 Day
Major	Acknowledgement within the day and analysis must be started within a day. Correction is expected within 2 days
Minor	Acknowledgement within 1 day and analysis must be started before the end of the sprint. Correction is expected by the end of the sprint.
Out of Scope	Acknowledgement within sprint and analysis must be completed before done the end of project lifespan. Correction is determined by stakeholders and backlog

7.6Metrics and Reporting

List the metrics being collected and reported during this test phase. Include reporting frequency, method, any tools used to gather and report the metrics and the intended audience.

Metric will be collected from the developer on this project; they will perform a stress test on the upload queue and stress test the OCR model. Performance will be presented to IT management and other stakeholders after the integration phase

7.7Escalation Path for Issues & Risks

If the project is utilizing a process other than the project Issues and Risk Management protocol, detail the leadership roles, stakeholders and other interested parties that must be notified of major issues or risks to this phase of testing.

N/A will be using Risk management protocol

8.Resources

Much of this information has been detailed in the Project's Test Strategy. Only list deviations from that information in the sections below (e.g. additional resources needed, deviations to the customary environments for this type of testing or exceptional test data needs).

8.1Test Estimation

Provide an estimate for the number of hours/cost/duration of testing activities associated to this test phase or include a link to a document that contains this information. If needed, describe the estimation strategy used: Story points, hours, retrospective adjustments, past estimates, duration, etc.

We estimate a total of 240 hours of work needed to complete this project. We estimate about 2 sprints need to be completed. One testing integration and one testing white box testing.

8.2Test Environments

Describe the test environments that will be utilized for this phase of testing. Include a description of measures and controls that will ensure the integrity of the test results.

All development will occur on local environments once Developers are able to complete development and integration the code will migrate into the user environment.

8.3Test Data Management

Describe the types of data required for the test and the approach that will be taken to obtain,

create or refresh the necessary data.

We will be using the current local environment of document creation as our initial data tracking how it performs when migrating to the cloud environment. Once that is complete the tester will move to new document creation and scanning in physical copies.

8.4Test Resources

Please provide a list of roles needed to complete this phase of testing and identify the individual who will be assigned to that role.

Role*	**Name**	**Role will be *i*nvolved with this Type of Testing (Duties)**
Developer	Justin	Responsible for black box testing local systems and integration testing
Quality Analyst	Jesse	Responsible for White box, integration testing, & Regression testing
Integration partner	Mark	Responsible for black box testing on partner backend

*Refer to the table of test roles defined in the Test Strategy document

9.Issues, Risks and Assumptions

Provide detail regarding assumptions, issues and risks that are associated with this test phase / product

increment. Projects can insert a link to other planning documents, tools etc. where this information is already being tracked.

9.1Assumptions for this Phase of Testing

List any assumptions/issues/risks that may apply during this test phase. These are testing specific and are in addition to overall project assumptions/risks/issues.

Assumption Number	Assumption
1	If cloud partner services go down then local storage would be the default for all new document creation storage
2	Existing Design of client interface will not have any changes
3	Permissions established in local environment will also apply to cloud environment

9.2Issues for this Phase of Testing

	Description		Responsible	Due Date

Issue #		Action	Resource	
1	Initial upload of all existing files cause a bottle neck and crashed local servers	Queue management timeout established	System Admins	11/01/2025
2	Changes do not save when user enters document after being sent to the cloud	Document history established	IT Developers	11/05/2025
3	OCR transcript is not being created	Misplaced file path for associated scanned document	IT Developers	11/10/2025
4	OCR is not handling non-english text	Initial machine learning did not have enough data for certain language	ML engineer	11/15/2025

9.3Risks for this Phase of Testing

Risk #	Description	Action	Responsible Resource	Due Date

1	Initial cloud storage allotment might be too low for initial testing	Contracts will be renegotiated for increased storage	Legal department	
2	Machine Learning Algorithm could take too long to train/ need more training time than available	OCR will launch but Machine learning aspect will release later	ML Engineers	
3	User will not be able to pull documents back down from the cloud	Project will be halted until fixed	Cloud service provider	
4	Scanned documents and transcripts do not link in upload process	Metadata will need to be re-engineered for proper linking	IT Developer	

10. Approvals

Approvers Signatures (must be authorized approvers/members of the project Change Control Board)

Role	Name/Title	Signature	Date
Business Owner (Authorized Approver)	Bob Bobington		
System Owner (if identified)	Cindy Cindyton		
Project Manager	Ross Rossington		
Project Test Manager	George Gerogington		

11.Appendix A – Example Defect Management Process for Waterfall Projects

The following table provides an example of the required steps of a “Defect Management Process”. Projects/Organizations may need to customize this approach to be specific to their environment and procedure.

Step	Role	Action to be Taken
1	Tester	Logs the defect into the defect tracking tool along with all pertinent details regarding how to recreate the defect. The tester assigns the defect(s) to the Developer Lead.
2	Developer Lead	Assigns the “NEW” defect to a developer and changes the status to “OPEN”.
3	Developer	Triages the “OPEN” defect. If the developer determines the defect is not a unique defect or cannot recreate the defect, the developer sets the status to “REJECT” and enters a reason code of “Cannot Duplicate”, “No Trouble Found” or “Duplicate”. If the Developer can confirm the defect and the defect needs to be corrected to comply with quality guidelines, the developer will correct the defect and set the status to “READY TO RETEST”. If the Developer can confirm the defect and the defect should not be corrected prior to release deployment, the Developer will set the status to “Out of Scope”.
4	Tester	Validates all “READY TO RETEST” defects. If the tester finds that the functionality is working fine; the defect status will be changed as “CLOSED”. If tester finds that the malfunction still exists, the tester will change the status of the defect to “REOPEN”. The Reopened defects will again go through the defect life cycle.
5	Tester	If a defect is set to the status of ‘REJECT’ by the developer, the Tester should review the defect and determine if they agree with the developer’s explanation as to why it is not a defect. If the Tester believes the defect should stand, it should be discussed in the daily triage meeting. If the tester agrees with the Rejected status, the Tester should close the defect.

6	**Developer Lead/Business Owner**	Any Defects with a status of “Out of Scope” are reviewed with the Business Owner prior to the Stage Exit. Deferred defects should be considered new items and prioritized within the current project processes for new item consideration.